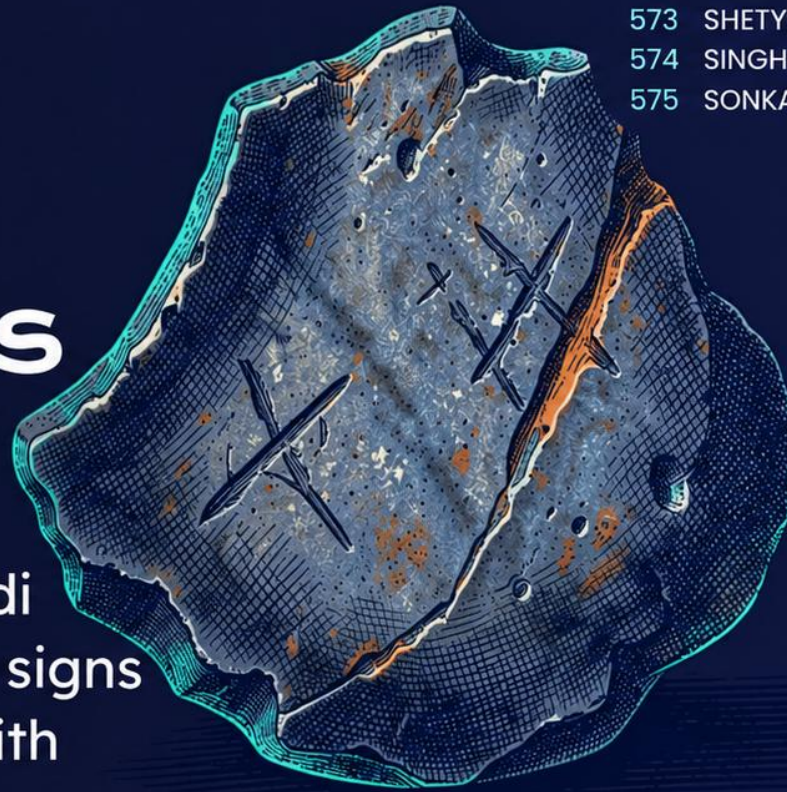


INDUS • KEELADI • VISUAL SIMILARITY

Can visual similarity connect Indus and Keeladi?

A CNN study tests whether Keeladi graffiti rank near expected Indus signs without confusing resemblance with linguistic descent.



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THE MOTIVATING COMPARISON

Four hand-matched signs are not enough

MATCH 01

I

MATCH 02

II

MATCH 03

III

MATCH 04

IV

The Keeladi corpus reports **1,001 graffiti sherds** and **56 Tamil-Brahmi inscribed sherds**, yet the motivating comparison rests on only four illustrated Indus matches.

Manual “looks similar” judgments do not scale, quantify uncertainty, or test whether random marks would also appear to match.

THE COMPUTATIONAL QUESTION

Which Indus sign does each Keeladi glyph rank highest, and is that ranking above chance?

The dataset is small, heterogeneous, and easy to mislabel

Dataset component	Images / classes · Role
Core Indus signs	880 / 45 folders · Training corpus
Matched Indus signs	100 / 5 classes · Known-pair context
Figure 65 real glyphs	39 / — · Digitised source glyphs
Keeladi validation	16 / 5 folders · Cross-domain evaluation
Tamil-Brahmi validation	58 / 3 groups · Out-of-domain control

43 model classes and 980 training images combine core Indus signs, matched signs, diacritics, modifiers, and 39 digitised Figure 65 glyphs. Validation includes 16 Keeladi images and 58 Tamil-Brahmi reference images, far below the reported archaeological corpus.

Three numbering systems coexist: Figure 65 serials, M-1977, and P-2010; allograph merging and annexure remapping are required before training.

The source glyphs were digitised from Figure 65 using vector table rules, avoiding table borders in the crops.

THE PIPELINE

Methodology: normalise, learn, audit

01

Digitise

Digitise real Figure 65 glyphs and harmonise labels.

02

Normalise

Grayscale, Otsu polarity, autocrop, deskew, 64×64 resizing, and [0,1] scaling.

03

Learn

Train a VGG-style CNN with a leakage-controlled split: divide originals first; augment training images only.

04

Audit

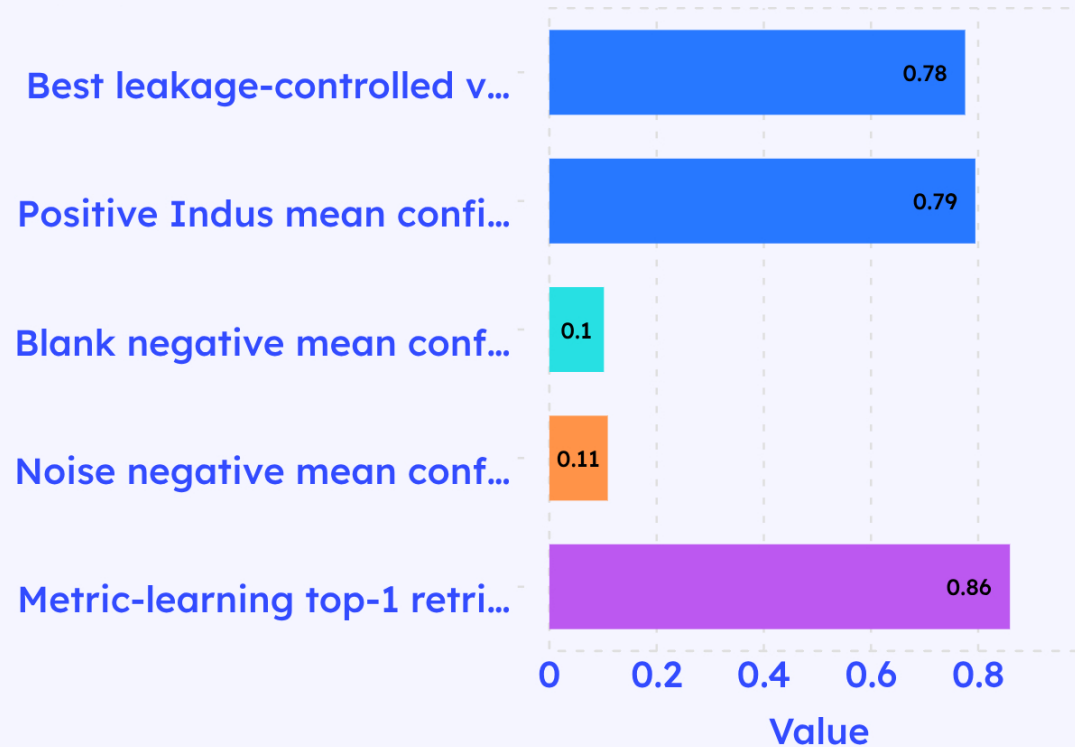
Audit leakage and open-set rejection; verify curated pairs with embeddings, stroke coverage, ranking, and permutation tests.

The model is a measurement instrument for visual similarity — not a decipherment system.

MEASURED RESULTS

The strongest result is methodological, not historical

Metric



HISTORICAL TEST

0 / 4

top-1 curated pairs

Verification improved the expected-score mean, but permutation $p = 0.1025$ remains non-significant.

Metric-learning retrieval reached 0.859 top-1 on perturbed Figure 65 glyphs, an upper bound rather than independent archaeological generalisation.

What the model still cannot claim



Visual similarity is not evidence of linguistic descent, decipherment, or a continuous script lineage.



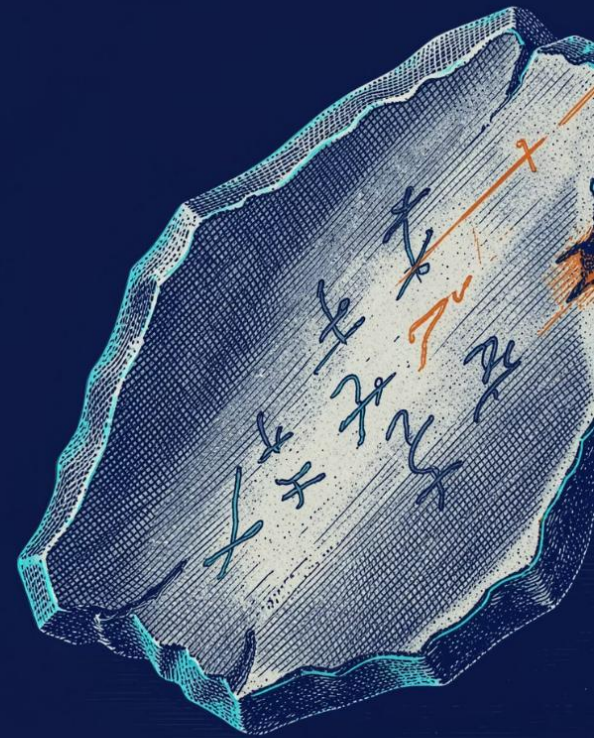
Stroke signs are intrinsically hard to separate. Several differ mainly by stroke count or position, and five signs scored 0.000 top-1.



Metric-learning queries are perturbations of source glyphs, so retrieval measures robustness, not independent allographs.



The corpus remains far below the 1,001-sherd research target. Domain adaptation and grid-decomposition features are not yet fully used.



DEFENSIBLE CLAIM

Conclusion: build the evidence before making the link

The project delivers a reproducible pipeline for digitisation, classification, open-set auditing, retrieval, and statistical verification.

After removing fabricated metrics and controlling leakage, the current corpus does not establish the four proposed Indus-Keeladi matches.

The next priority is real, labelled data: expand Indus allographs and Keeladi coverage, add real negatives, calibrate confidence, and rerun the audit.

The defensible claim today is narrow: the pipeline can measure visual correspondence; it cannot establish script descent.

